

Summary of EH chapter 3

- can introduce complex shifted momenta $\hat{p}_i = p_i + z \epsilon_i$ st. momentum conservation holds and all are on shell $\hat{p}_i^2 = 0$.

- consider $\hat{A}_n(z)$ - holomorphic in z
 - no branch cuts
 - only has simple poles } Tree level

- the real amplitude is $A_n = \hat{A}_n(0) = - \sum_{\text{poles}} \text{Res} [\hat{A}_n(z)/z] + B_n$

- the boundary term is the $O(z^0)$ term in the $z \rightarrow \infty$ limit of $A_n - B_n = 0$ is a requirement for many recursion relations.

- at the poles in z the amplitude factorizes and if $B_n = 0$, the amplitude is determined by its poles:

$$A_n = \sum_{\text{diagrams } I} \hat{A}_L(z_I) \frac{1}{P_I^2} \hat{A}_R(z_I)$$

this is the general recursion formula.

- BCFW: choose Γ_i and Γ_j st. $|\hat{i}\rangle = |i\rangle + z|j\rangle$, $|\hat{j}\rangle = |j\rangle + z|i\rangle$ and no other spinors shifted. and use the above recursion formula

- gluon amplitudes and B_n :

$$\begin{array}{ccccc} [i, j] \text{-shift} & [-, -] & [-, +] & [+, +] & [+, -] \\ \hat{A}_n(z) \sim & z^{-1} & z^{-1} & z^{-1} & z^{-3} \end{array}$$

here $i \nleftrightarrow j$ are adjacent. the amp is suppressed by an add power of z^{-1} if $i \nleftrightarrow j$ are non-adjacent.

Problems: 3.1, 3.2, 3.5, 3.6 + Prove Parke-Taylor using non-adj shifts

3.1 $-P = |p\rangle [p] + |p\rangle \langle p|$

$$\hat{i} = i + z j \quad \hat{j} = j \quad \hat{i} = i \quad |j\rangle = j - z |i\rangle$$

$$\begin{aligned} -\hat{P} &= -P + z P \\ &= |p\rangle \underbrace{[p] + |p\rangle \langle p|}_{-P} + z (|i\rangle [i] + |i\rangle \langle i|) \end{aligned}$$

$$\begin{aligned} -\hat{P} &= \hat{i} \langle i + z j \rangle + \hat{i} \langle i \\ &= i \langle i + z j \rangle + i \langle i + z i \rangle j \\ &= -P + z (j \langle i + i \rangle j) \end{aligned}$$

$$-\hat{P} = -P - z (i \langle j + j \rangle i)$$

choose Γ_i st. $\hat{P}_i = j \langle i + i \rangle j$
 Γ_j st. $\hat{P}_j = -i \langle j + j \rangle i$

(recall that the λ & $\tilde{\lambda}$ of P are eval for complex momenta)

now, we want to extract the μ components from $\hat{P}_{i,j}$

$$\Gamma_{a\dot{a}} = \Gamma_\mu \sigma^\mu_{a\dot{a}} \quad \text{and} \quad \Gamma_\mu = -\frac{1}{2} \Gamma_{a\dot{a}} \bar{\sigma}_\mu^{\dot{a}a}$$

3.2 $A_n [1^- 2^- \dots n^+] = \frac{\langle 12 \rangle^4}{\langle 12 \rangle \langle 23 \rangle \dots \langle n1 \rangle}$

$$\begin{aligned} |\hat{1}\rangle &= |1\rangle + z |2\rangle & |\hat{1}\rangle &= |1\rangle \\ |\hat{2}\rangle &= |2\rangle & |\hat{2}\rangle &= |2\rangle - z |1\rangle \end{aligned}$$

$$\begin{aligned} \langle 12 \rangle &\rightarrow \langle \hat{1}\hat{2} \rangle = \langle 12 \rangle \\ \langle 1k \rangle &\rightarrow \langle \hat{1}k \rangle = \langle 1k \rangle & k=3, \dots, n \\ \langle 2k \rangle &\rightarrow \langle \hat{2}k \rangle = \langle 2k \rangle - z \langle 1k \rangle & k=3, \dots, n \end{aligned}$$

$$\hat{A}_n = \frac{\langle 12 \rangle^4}{\langle 12 \rangle (\langle 23 \rangle - z \langle 13 \rangle) \langle 34 \rangle \dots \langle n1 \rangle} \sim \frac{1}{z} \text{ for large } z$$

Other shifts:

1) $[1, k]$ for $k = 3, \dots, n-1$

$$\begin{aligned} |\hat{1}\rangle &= |1\rangle + z |k\rangle & |\hat{1}\rangle &= |1\rangle \\ |\hat{k}\rangle &= |k\rangle & |\hat{k}\rangle &= |k\rangle - z |1\rangle \end{aligned}$$

$$\langle \hat{k}j \rangle = \langle kj \rangle - z \langle 1j \rangle \text{ for any } j$$

$\uparrow$
appears 2x in denom $\Rightarrow \hat{A}_n \sim 1/z^2$ at large z

2) $[k, 1]$ for $k=3, \dots, n-1$

$$\begin{aligned} |\hat{k}\rangle &= |k\rangle + z |1\rangle & |\hat{k}\rangle &= |k\rangle \\ |1\rangle &= |1\rangle & |1\rangle &= |1\rangle - z |k\rangle \end{aligned}$$

$$\begin{aligned} \langle \hat{k} | \hat{j} \rangle &= \langle k | j \rangle \\ \langle \hat{1} | \hat{j} \rangle &= \langle 1 | j \rangle - z \langle k | j \rangle \end{aligned}$$

$$\begin{aligned} \hat{A}_n &\rightarrow \frac{\langle \hat{1} \hat{2} \rangle^4}{\langle \hat{1} \hat{2} \rangle \langle \hat{2} \hat{3} \rangle \dots \langle \hat{k-1}, \hat{k} \rangle \langle \hat{k}, \hat{k+1} \rangle \dots \langle \hat{n} \hat{1} \rangle} \\ &= \frac{(\langle 12 \rangle - z \langle k2 \rangle)^3}{\langle 23 \rangle \dots \langle n1 \rangle - z \langle kn \rangle)} \sim z^2 \text{ at large } z. \end{aligned}$$

3) $[k, j]$ adj & non-adj where $k, j \neq 1, 2$

$$\begin{aligned} |\hat{k}\rangle &= |k\rangle + z |j\rangle & |\hat{j}\rangle &= |j\rangle \\ |\hat{k}\rangle &= |k\rangle & |\hat{j}\rangle &= |j\rangle - z |k\rangle \end{aligned}$$

$$\begin{aligned} \langle \hat{k} | \hat{j} \rangle &= \langle k | j \rangle \\ \langle \hat{k} | i \rangle &= \langle k | i \rangle \text{ for } i \neq j, 1, 2 \\ \langle \hat{j} | i \rangle &= \langle j | i \rangle - z \langle k | i \rangle \text{ for } i \neq k, j, 1, 2 \end{aligned}$$

$$\begin{aligned} \hat{A}_n &\rightarrow \frac{\langle 12 \rangle^4}{\langle 12 \rangle \langle 23 \rangle \dots \langle k-1, \hat{k} \rangle \langle \hat{k}, k+1 \rangle \dots \langle j-1, \hat{j} \rangle \langle \hat{j}, j+1 \rangle \dots \langle n1 \rangle} \\ &= \frac{\langle 12 \rangle^4}{\langle 12 \rangle \langle 23 \rangle \dots \langle k-1, k \rangle \langle k, k+1 \rangle \dots (\langle j-1, j \rangle - z \langle j-1, k \rangle) (\langle j, j+1 \rangle - z \langle k, j+1 \rangle) \dots \langle n1 \rangle} \\ &\sim \frac{1}{z^2} \text{ for large } z \end{aligned}$$

if $k \neq j$ were adj the $\langle k | j \rangle$ bracket would remain unshifted
so $A_n \sim z^2$ for large z .

3.5 $M_4(1^- 2^- 3^+ 4^+) = z$

$$\begin{aligned} M_3(1^- 2^- 3^+) &= \frac{\langle 12 \rangle^6}{\langle 13 \rangle^2 \langle 23 \rangle^2} \\ M_3[1^+ 2^+ 3^-] &= \frac{[12]^6}{[13]^2 [23]^2} \end{aligned}$$

$[1, 2]$ - shift.

$$\begin{aligned} \hat{1} &= 1 + z 2 \\ \hat{2} &= 2 \\ \hat{1} &= 1 \\ \hat{2} &= 2 - z 1 \end{aligned}$$

$$\begin{aligned} M_4[1^- 2^- 3^+ 4^+] &= \text{Diagram: } \begin{array}{c} \hat{1}^- \\ | \\ \textcircled{L} \text{---} \textcircled{R} \\ | \quad | \\ 4^+ \quad 3^+ \end{array} \text{ with } \hat{P}_2 \text{ above the line} + (3 \leftrightarrow 4) \\ &= \hat{M}_3[\hat{1}^- \hat{P}_2^+ 4^+] \frac{1}{P_1^2} \hat{M}_3[-\hat{P}_1^+ \hat{2}^- 3^+] \quad \leftarrow \text{Sum over } h_I + (3 \leftrightarrow 4) \end{aligned}$$

$$\hat{P}_I = -\hat{P}_{14} \rightarrow \text{on shell cond. } \hat{P}_{14}^2 = 2 \hat{p}_1 \cdot p_4 = \langle \hat{1}4 \rangle [\hat{1}4] = \langle 14 \rangle [14] = 0$$

$$\Rightarrow [\hat{1}4] = 0 \text{ for generic momenta}$$

$$|\hat{P}_{14}\rangle [\hat{P}_{14} 4] = -\hat{P}_{14} |4\rangle = -(\hat{p}_1 + p_4) |4\rangle = -\hat{p}_1 |4\rangle = |\hat{1}\rangle [\hat{1}4] = 0$$

$$\Rightarrow [\hat{P}_{14} 4] = 0 \text{ since } |\hat{P}_{14}\rangle \neq 0$$

$$\hat{P}_I = \hat{P}_{23} \rightarrow \hat{P}_{23}^2 = 2 \hat{p}_2 \cdot p_3 = \langle \hat{2}3 \rangle [\hat{2}3] = \langle 23 \rangle [23] = 0$$

$$\Rightarrow \langle \hat{2}3 \rangle = 0 \text{ for general momenta}$$

$$\Rightarrow \langle \hat{P}_{23} 2 \rangle = 0 \text{ by a similar argument to that above.}$$

$$\text{when } h_I = + \text{ then } \hat{M}_3 [\hat{1}^- \hat{P}_{14}^+ 4^+] = \left(\frac{[P_{14} 4]^3}{[P_{14} \hat{1}] [4 \hat{1}]} \right)^2 \rightarrow 0$$

$$\hat{M}_3 [-\hat{P}_{23}^- \hat{2}^- 3^+] = \left(\frac{\langle \hat{P}_{23} \hat{2} \rangle^3}{\langle \hat{P}_{23} \rangle \langle \hat{2}3 \rangle} \right)^2 \rightarrow 0$$

$$M_4 [1^- 2^- 3^+ 4^+] = \hat{M}_3 [\hat{1}^- \hat{P}_{14}^- 4^+] \frac{1}{\hat{P}_I^2} \hat{M}_3 [\hat{P}_{14}^+ \hat{2}^- 3^+] + (3 \leftrightarrow 4)$$

$$= \left(\frac{\langle \hat{1} \hat{P}_{14} \rangle^3}{\langle \hat{1}4 \rangle \langle \hat{P}_{14} 4 \rangle} \right)^2 \frac{1}{\langle 14 \rangle [14]} \left(\frac{[\hat{P}_{14} 3]^3}{[\hat{P}_{14} \hat{2}] [3 \hat{2}]} \right)^2 + (3 \leftrightarrow 4)$$

$$\begin{aligned} \langle \hat{1} \hat{P}_{14} \rangle [\hat{P}_{14} 3] &= -\langle \hat{1} | \hat{P}_{14} | 3 \rangle \\ &= -\langle \hat{1} | \hat{1} + 4 | 3 \rangle \\ &= -\langle \hat{1} 4 | 3 \rangle \\ &= -\langle 14 | 3 \rangle \\ &= -\langle 14 \rangle [43] \end{aligned}$$

$$\begin{aligned} \langle \hat{P}_{14} 4 \rangle [\hat{P}_{14} \hat{2}] &= -\langle 4 | \hat{P}_{14} | \hat{2} \rangle \\ &= -\langle 4 | \hat{1} + 4 | \hat{2} \rangle \\ &= -\langle 4 | \hat{1} | \hat{2} \rangle \\ &= +\langle 4 \hat{1} \rangle [\hat{1} \hat{2}] \\ &= \langle 41 \rangle [12] \\ &= -\langle 14 \rangle [12] \end{aligned}$$

$$M_4 = \frac{(-\langle 14 \rangle [43])^6}{\langle \hat{1}4 \rangle^2 (-\langle 14 \rangle [12])^2 [3 \hat{2}]^2 \langle 14 \rangle [14]} + (3 \leftrightarrow 4)$$

$$= \frac{\langle 14 \rangle^6 [13]^6}{\langle 14 \rangle^5 [12]^2 [32]^2 [14]} + (3 \leftrightarrow 4)$$

$$= \frac{[43]^6}{[12]^2 [32]^2} \frac{\langle 14 \rangle}{[14]} + (3 \leftrightarrow 4)$$

$$= \frac{[43]^6}{[12]^2 [23]^2} \frac{\langle 14 \rangle}{[14]} + \frac{[34]^6}{[12]^2 [24]^2} \frac{\langle 13 \rangle}{[13]} \quad -S_{12} = [12] \langle 12 \rangle$$

$$= \frac{[34]^6}{[12]^2} \left(\frac{\langle 14 \rangle}{[14] [23]^2} + \frac{\langle 13 \rangle}{[24]^2 [13]} \right)$$

$$= \frac{[34]^6}{[12]} \frac{\langle 12 \rangle}{[14] [23] [3] [24]} \left(\frac{\langle 14 \rangle [24] [13]}{\langle 12 \rangle [23] [12]} + \frac{\langle 13 \rangle [14] [23]}{\langle 12 \rangle [24] [12]} \right)$$

$$-S_{12} A_4 [1^- 2^- 3^+ 4^+] A_4 [1^- 2^- 4^+ 3^+]$$

Parke-Taylor.

(X)

$$\begin{aligned}
M_4[1^- 2^- 3^+ 4^+] &= -S_2 A_4 [1^- 3^+ 4^+] A_4 [1^- 2^+ 3^+] \\
&= -\langle 12 \rangle [12] \frac{[34]^4}{[34][41][12][23]} \frac{[34]^4}{[43][31][12][24]} \\
&= +\langle 12 \rangle [12] \frac{[34]^8}{[34]^2 [12]^2 [41][23][31][24]} \\
&= \frac{\langle 12 \rangle [34]^6}{[12] [14][23][13][24]} \\
&= \frac{[34]^6}{[12]} \left(\frac{\langle 12 \rangle}{[14][23][13][24]} \right)
\end{aligned}$$

We want to show that $\otimes = 1$

$$\begin{aligned}
\otimes &= \left(\frac{\langle 14 \rangle [24][13]}{\langle 12 \rangle [23][12]} + \frac{\langle 13 \rangle [14][23]}{\langle 12 \rangle [24][12]} \right) \\
&= \frac{\langle 14 \rangle [24][13]}{\langle 14 \rangle [34][12]} - \frac{\langle 13 \rangle [14][23]}{\langle 13 \rangle [34][12]} \\
&= \frac{[24][13] - [14][23]}{[34][12]} \\
&= \frac{[12][34]}{[34][12]} \\
&= 1
\end{aligned}$$

$$1) [1+2][2+3][3+4][4] = 0$$

$$\langle 12 \rangle [23] = -\langle 14 \rangle [43]$$

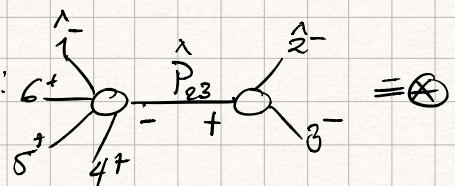
$$\langle 12 \rangle [24] = -\langle 13 \rangle [34]$$

$$[12][34] + [13][42] + [14][23] = 0$$

$$\Rightarrow -[13][24] + [14][23] = -[12][34]$$

$$\Rightarrow [13][24] - [14][23] = [12][34]$$

3.6 we are to show that the diagram :



does not contribute to $A_6[1^- 2^- 3^- 4^+ 5^+ 6^+]$ with a $[1, 2]$ shift

$$\begin{aligned}
\otimes &\propto A_3[\hat{P}_{23}^+, \hat{2}^-, 3^-] \quad \langle \hat{2} \hat{P}_{23} \rangle \langle 3 \hat{P}_{23} \rangle \\
&= \frac{\langle \hat{2} 3 \rangle^3}{\langle \hat{2} \hat{P}_{23} \rangle \langle 3 \hat{P}_{23} \rangle}
\end{aligned}$$

on-shell cond: $\hat{P}_{23}^2 = 0 = \langle \hat{2} 3 \rangle [\hat{2} 3] = \langle \hat{2} 3 \rangle [23] = 0 \Rightarrow \langle \hat{2} 3 \rangle = 0$ since for general momenta $[23] \neq 0$

$$\begin{aligned}
[\hat{P}_{23}] \langle \hat{P}_{23} | \hat{2} \rangle &= -\hat{P}_{23} | \hat{2} \rangle = -(\hat{p}_2 + \hat{p}_3) | \hat{2} \rangle = -\hat{p}_3 | \hat{2} \rangle = [13] \langle 3 \hat{2} \rangle = 0 \\
&\Rightarrow \langle \hat{P}_{23} \hat{2} \rangle = 0 \quad \text{as } |\hat{P}_{23}\rangle \neq 0 \text{ for general momenta}
\end{aligned}$$

$$\begin{aligned}
[\hat{P}_{23}] \langle \hat{P}_{23} | 3 \rangle &= -\hat{P}_{23} | 3 \rangle = -\hat{p}_2 | 3 \rangle = -|\hat{2}\rangle \langle \hat{2} 3 \rangle = 0 \\
&\Rightarrow \langle \hat{P}_{23} | 3 \rangle = 0 \quad \text{as } | \hat{2} \rangle \neq 0 \text{ in general}
\end{aligned}$$

So $A_3 \sim \frac{0^3}{0^2} \sim 0$ and the P_{23} channel does not contribute.

Pocke-Taylor for non-adj shifts

[1, 3⁺] shift: $\begin{aligned} \hat{1}^- &= 1^- + z 13^- \\ \hat{3}^- &= 3^- \end{aligned} \quad \begin{aligned} \hat{1}^+ &= 1^+ \\ \hat{3}^+ &= 3^+ - z 11^+ \end{aligned}$

$$A_n[1^- 2^- 3^+ \dots n^+] = \sum_{k=5}^n \left(\text{diagram 1} \right) + \sum_{k=4}^n \left(\text{diagram 2} \right)$$

only 3pt
Amp. is non-zero
 $\Rightarrow = 5$

non-zero if $k=n$

non-zero
 $\neq k=4$

$$\begin{aligned} &= \hat{A}_{n-2}[\hat{1}^- z \hat{P}_{34}^+ 5^+ \dots n^+] \frac{1}{P_{34}^2} \hat{A}_3[-\hat{P}_{34}^-, \hat{3}^+ 4^+] \\ &+ \hat{A}_3[\hat{1}^-, -\hat{P}_m^+, n^+] \frac{1}{P_m^2} \hat{A}_{n-2}[\hat{P}_m^-, 2^-, \hat{3}^+, \dots, n^{+1}] \\ &+ \hat{A}_{n-2}[\hat{1}^- \hat{P}_{23}^- 4^+ \dots n^+] \frac{1}{P_{23}^2} A_3[-\hat{P}_{23}^+, 2^-, \hat{3}^+] \end{aligned}$$

diagram 2: the only diagram when $\hat{P}_m^2$ gives a [] constraint which is what we need to see if one of the entire MHV A_3 's are zero.

$$\hat{P}_m^2 = 0 = \langle \hat{1} n \rangle [\hat{1} n] = \langle m n \rangle [\hat{1} n] \Rightarrow [\hat{1} n] = 0$$

$$\hat{A}_3[\hat{1}^-, -\hat{P}_m^+, n^+] = \frac{[\hat{P}_m n]^4}{[\hat{P}_m \hat{1}][n \hat{1}]}$$

$$|\hat{P}_m\rangle [\hat{P}_m n] = -\hat{P}_m |n\rangle = -\hat{P}_1 |n\rangle = |1\rangle [\hat{1} n] = 0$$

$$\Rightarrow [\hat{P}_m n] = 0$$

$$\Rightarrow \hat{A}_3[\hat{1}^-, -\hat{P}_m^+, n^+] \rightarrow 0.$$

$$\begin{aligned} A_n[1^- 2^- 3^+ \dots n^+] &= \hat{A}_{n-2}[\hat{1}^- z \hat{P}_{34}^+ 5^+ \dots n^+] \frac{1}{P_{34}^2} \hat{A}_3[-\hat{P}_{34}^-, \hat{3}^+ 4^+] \leftarrow \textcircled{1} \\ &+ \hat{A}_{n-2}[\hat{1}^- \hat{P}_{23}^- 4^+ \dots n^+] \frac{1}{P_{23}^2} A_3[-\hat{P}_{23}^+, 2^-, \hat{3}^+] \leftarrow \textcircled{2} \end{aligned}$$

$$\begin{aligned} \textcircled{1} &= \frac{\langle \hat{1} 2 \rangle^4}{\langle \hat{1} 2 \rangle \langle 2 \hat{P}_{34} \rangle \langle \hat{P}_{34} 5 \rangle \langle 5 6 \rangle \dots \langle n \hat{1} \rangle} \frac{1}{\langle 3 4 \rangle [3 4]} \frac{[\hat{3} 4]^3}{[\hat{3} \hat{P}_{34}] [4 \hat{P}_{34}]} \\ &= \frac{\langle \hat{1} 2 \rangle^4}{\langle \hat{1} 2 \rangle \langle 2 \hat{3} \rangle [4 3] \langle 5 4 \rangle [4 3] \langle 5 6 \rangle \dots \langle n \hat{1} \rangle} \frac{1}{\langle 3 4 \rangle [3 4]} \\ &= \frac{\langle \hat{1} 2 \rangle^4}{\langle \hat{1} 2 \rangle \langle 2 3 \rangle \langle 3 4 \rangle \langle 4 5 \rangle \dots \langle n 1 \rangle} \left(\frac{\langle 3 4 \rangle}{\langle 2 3 \rangle} \right) \end{aligned}$$

$$\begin{aligned} \langle 2 \hat{P}_{34} \rangle [4 \hat{P}_{34}] &= \langle 2 | \hat{P}_{34} | 4 \rangle = \langle 2 \hat{3} \rangle [3 4] \\ \langle \hat{P}_{34} 5 \rangle [\hat{3} \hat{P}_{34}] &= -\langle 5 | \hat{P}_{34} | \hat{3} \rangle = \langle 5 4 \rangle [4 3] \end{aligned}$$

$$\begin{aligned} \textcircled{2} &= \frac{\langle \hat{1} \hat{P}_{23} \rangle^{43}}{\langle \hat{1} \hat{P}_{23} \rangle \langle \hat{P}_{23} 4 \rangle \langle 4 5 \rangle \dots \langle n \hat{1} \rangle} \frac{1}{\langle 2 3 \rangle [2 3]} \frac{[\hat{P}_{23} \hat{3}]^3}{[\hat{P}_{23} 2] [3 2]} \\ &= \frac{\langle \hat{1} 2 \rangle^3 [2 3]^3}{\langle \hat{3} 4 \rangle [3 2] \langle 4 5 \rangle \dots \langle n 1 \rangle \langle 2 3 \rangle [2 3] [3 2]} \\ &= \frac{\langle \hat{1} 2 \rangle^4}{\langle \hat{1} 2 \rangle \langle 2 3 \rangle \langle 3 4 \rangle \langle 4 5 \rangle \dots \langle n 1 \rangle} \left(\frac{\langle 3 4 \rangle}{\langle \hat{3} 4 \rangle} \right) \end{aligned}$$

$$\begin{aligned} \langle \hat{1} \hat{P}_{23} \rangle [\hat{P}_{23} \hat{3}] &= -\langle \hat{1} | \hat{P}_{23} | \hat{3} \rangle = -\langle \hat{1} 2 | \hat{3} \rangle = +\langle \hat{1} 2 \rangle [2 \hat{3}] = \langle \hat{1} 2 \rangle [2 3] \\ \langle \hat{P}_{23} 4 \rangle [\hat{P}_{23} 2] &= +\langle 4 | \hat{P}_{23} | 2 \rangle = -\langle 4 \hat{3} \rangle [\hat{3} 2] = -\langle \hat{3} 4 \rangle [2 3] \end{aligned}$$

$$A_n = \textcircled{1} + \textcircled{2} = \frac{\langle \hat{1} 2 \rangle^4}{\langle \hat{1} 2 \rangle \langle 2 3 \rangle \dots \langle n 1 \rangle} \left(\underbrace{\frac{\langle 3 4 \rangle}{\langle \hat{3} 4 \rangle} + \frac{\langle 2 3 \rangle}{\langle 2 \hat{3} \rangle}}_{\textcircled{*} \text{ must be 1}} \right)$$

$$\begin{aligned} \hat{P}_{23}^2 = 0 &= \underbrace{\langle 2 \hat{3} \rangle}_{=0} [2 3] \\ \langle 2 \hat{3} \rangle &= \langle 2 3 \rangle - Z_{21} \langle 2 1 \rangle = 0 \Rightarrow Z_{23} = \langle 2 3 \rangle / \langle 2 1 \rangle \\ \langle \hat{3} 4 \rangle &= \langle 3 4 \rangle - Z_{23} \langle 1 4 \rangle = \langle 3 4 \rangle - \frac{\langle 2 3 \rangle}{\langle 2 1 \rangle} \langle 1 4 \rangle \end{aligned} \quad \left. \vphantom{\begin{aligned} \hat{P}_{23}^2 = 0 \\ \langle 2 \hat{3} \rangle \\ \langle \hat{3} 4 \rangle \end{aligned}} \right\} \text{for } \textcircled{2}$$

$$\begin{aligned} \hat{P}_{34}^2 = 0 &= \langle \hat{3} 4 \rangle [\hat{3} 4] = \underbrace{\langle \hat{3} 4 \rangle}_{=0} [3 4] \\ \langle \hat{3} 4 \rangle &= \langle 3 4 \rangle - Z_{34} \langle 1 4 \rangle = 0 \Rightarrow Z_{34} = \langle 3 4 \rangle / \langle 1 4 \rangle \\ \langle 2 \hat{3} \rangle &= \langle 2 3 \rangle - Z_{34} \langle 2 1 \rangle = \langle 2 3 \rangle - \frac{\langle 3 4 \rangle}{\langle 1 4 \rangle} \langle 2 1 \rangle \end{aligned} \quad \left. \vphantom{\begin{aligned} \hat{P}_{34}^2 = 0 \\ \langle \hat{3} 4 \rangle \\ \langle 2 \hat{3} \rangle \end{aligned}} \right\} \text{for } \textcircled{1}$$

$$\begin{aligned} \textcircled{*} &= \frac{\langle 3 4 \rangle}{\langle \hat{3} 4 \rangle} + \frac{\langle 2 3 \rangle}{\langle 2 \hat{3} \rangle} \\ &= \frac{\langle 3 4 \rangle}{\langle 3 4 \rangle - \frac{\langle 2 3 \rangle}{\langle 1 4 \rangle} \langle 1 4 \rangle} + \frac{\langle 2 3 \rangle}{\langle 2 3 \rangle - \frac{\langle 3 4 \rangle}{\langle 1 4 \rangle} \langle 2 1 \rangle} \\ &= \frac{\langle 3 4 \rangle \langle 1 4 \rangle}{\langle 1 4 \rangle \langle 3 4 \rangle - \langle 2 3 \rangle \langle 1 4 \rangle} + \frac{\langle 2 3 \rangle \langle 1 4 \rangle}{\langle 2 3 \rangle \langle 1 4 \rangle - \langle 3 4 \rangle \langle 2 1 \rangle} \\ &= 1 \end{aligned}$$